

Multimodal AI for Human Sensing and Interaction

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Abstract

A significant body of HCI research today focuses on applying AI to sense, learn, and interact with humans through a wide range of wearable and ubiquitous sensors. These methods typically involve learning features from multimodal sensory data using AI methods. To aid HCI researchers who want to apply AI to their sensing problems, this course will cover the fundamental challenges and approaches in multimodal AI for human sensing and interaction. It is planned for 3 parts, one given by each organizer. The first covers the foundations of multimodal AI, studying how AI systems can represent, combine, and learn information from many interconnected sensory inputs. The second part discusses the practice of multimodal AI for human sensing, covering the latest methods for cross-modal learning across diverse sensors, human-centered application domains, and real-world concerns around their usage. The final part covers the hardware, fabrication, and data collection challenges that must be tackled to deploy these multimodal AI systems in the real world. By the end of this course, attendees should understand the fundamental principles and challenges of multimodal AI, identify the right AI approaches for their problems, prototype basic hardware systems for efficient and robust sensing, be aware of real-world concerns around ethics, interpretability, and privacy, and appreciate the range of human-centered applications enabled by multimodal AI and sensing.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interaction techniques**; • **Computing methodologies** → **Machine learning; Artificial intelligence**.

Keywords

AI, machine learning, generative AI, multimodal AI, AI and sensing, human-AI interaction, health and wellness

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1 Introduction

Humans rely on multimodal sensory inputs, including visual, auditory, and tactile information, to navigate and interpret the world around them. The capturing, understanding, and generation of multimodal information are essential for creating intelligent systems capable of understanding and responding to complex human behaviors and environments, ultimately driving innovation across a wide range of applications, including healthcare, education, robotics, and human-computer interactions. Today, there is significant interest in the HCI community in developing sensors and AI models to process these multimodal sensory inputs for human sensing and interaction. This course is designed to give an overview of the field of multimodal AI for human sensing. It is planned for 3 parts, one given by each of the organizers.

The first covers the foundations of multimodal AI, studying how AI systems can represent, combine, and learn information from many interconnected sensory inputs [7]. We will start by explaining the three key principles driving multimodal challenges and innovations: modality heterogeneity, connections, and interactions [5]. Building upon these definitions, we propose a new taxonomy of six core challenges in multimodal learning: representation, alignment, reasoning, generation, transference, and quantification. Finally, we will end with recent advances in this fast-growing field, including multimodal transformers [13], foundation models [6], generative AI, and interactive agents.

The second part of the course will cover the practice of AI for human sensing. Our focus will begin with cross-modal learning, a critical component enabling AI systems to synthesize and augment diverse sensor data types. We will analyze techniques for transforming abundant and well-labeled video data into modalities for which labeled data is not readily available such as Doppler, IMU, and low-fi audio [1]. Given the close nature of human sensing, we will also cover the critical deployment concerns of privacy, evaluate error handling and uncertainty quantification, examine fairness challenges, and discuss the integration of human-in-the-loop methodologies to build systems that augment human decision-making and agency [10]. Finally, we will cover real-world applications including human activity recognition, pose estimation, gesture recognition, and sensing for infrastructure, environment, wellness, and health monitoring [2, 11].

The third part of the course will discuss hardware solutions designed for multimodal data collection, extending beyond traditional modalities like visual, audio, and text to include physical interaction sensing [8, 9]. It will cover the prototyping of hardware systems, data acquisition, and multisensory fusion for capturing proprioception, tactile sensing, and muscle activity [4]. We will also highlight the challenges of acquiring diverse data through hardware platforms and the technical considerations required for accurate,

synchronized sensing [12]. Furthermore, the session will explore the potential of multisensor fusion systems [3], which integrate data from multiple sensing modalities to create richer, more robust representations of human interactions. We will discuss how these integrated hardware solutions and fusion techniques enable more comprehensive and intelligent sensing of physical interactions and behaviors.

This course is related to these hot topics: (1) Big Data, Machine Learning, and AI for Research/Design, (2) Generative AI (LLMs, VLMs, diffusion models) for HCI, (3) Intelligent and Autonomous Systems, (4) Human-AI Interaction, (5) HCI for Health, (6) Risks and Bias in AI (Responsible AI, AI ethics, etc), (7) Data Science, (8) Tools, Skills, and Methodologies of interest to the CHI Audience.

2 Benefits and learning objectives

This course will equip attendees with the following:

- Fundamental principles of multimodal heterogeneity, connections, and interactions, to know when to use and collect multimodal data, and when adding a modality could cause harms (e.g., worse performance, optimization challenges, higher complexity).
- Technical challenges and advances: Given a multimodal task of interest, how do we decompose it into the technical challenges of representation, alignment, reasoning, generation, transference, and quantification? What are the latest models in each area, and which can be used for their use case? What remains to be solved in each technical challenge, and how can we design principled approaches towards tackling them?
- Cross-modal learning, to transform abundant and well-labeled video data into modalities for which labeled data is not readily available such as Doppler, IMU, and low-fi audio for accurate human sensing.
- Awareness of the hardware challenges for multimodal data collection and physical interaction sensing, and understanding how to prototype hardware systems for efficient and robust sensing.
- Real-world concerns around ethics, interpretability, and privacy in AI systems, ways to assess privacy concerns, evaluate error handling and uncertainty quantification strategies, examine fairness challenges, and integrate human-in-the-loop methodologies to build systems that augment human decision-making and agency.
- Appreciate the range of applications enabled by multimodal AI and sensing, including human activity recognition, pose estimation, gesture recognition, and sensing technologies for infrastructure, environment, wellness, and health monitoring, among others.

3 Intended audience

The course is intended for junior and senior researchers in HCI interested in learning about AI and sensing. These can be relevant to many practitioners including in healthcare, robotics, multimodal interaction, multimedia, and AR/VR.

4 Prerequisites

We expect the audience to have an introductory computer science, AI, or HCI background.

5 Content

The following is a detailed schedule and list of topics:

Part 1: Foundations of Multimodal AI (Paul, 45 mins)

Introduction (15 mins)

- What is multimodal? Historical view and multimodal research tasks.
- Key principles of multimodal: heterogeneity, connections, interactions.

Multimodal challenges (15 mins)

- Why is multimodal hard? Introduction to core challenges.
- Overview of multimodal representation, alignment, reasoning, transfer, generation, and quantification.
- Identifying recent solutions for practitioners.

Recent advances in multimodal AI (15 mins)

- Multimodal transformers and foundation models
- Multimodal generative models
- Multimodal agents

Part 2: Multimodal AI for Human Sensing (Karan, 45 mins)

Cross-modal learning (20 mins)

- Sensor data synthesis: Video to Doppler, Video to IMU, Video to Audio, MoCap to IMU, MoCap to UWB
- Data augmentation
- Temporal data modeling

Ethics, interpretability and privacy (10 mins)

- Privacy and fairness concerns
- Handling errors and uncertainty
- Bringing humans into the loop

Applications (15 mins)

- Human activity recognition, pose estimation, gesture recognition
- Infrastructure and environmental sensing
- Wellness and fitness tracking, mobile health monitoring

Part 3: Hardware and Sensors for Multimodal AI (Yiyue, 45 mins)

Introduction (10 mins)

- Challenges and opportunities in hardware and sensors for multimodal AI
- Importance of scalable, customizable hardware platforms
- Key applications benefiting from advancements in multimodal sensing and feedback interfaces

Multimodal sensing and feedback interface (20 mins)

- Advanced fabrication techniques for multimodal sensing hardware
- Addressing scalability, adaptability, and customization in hardware development
- Innovations in state-of-the-art data acquisition systems for multimodal interfaces
- Integration of diverse sensing modalities into compact, flexible form factors

Multisensory data fusion (15 mins)

- Approaches to synchronize and interpret multimodal data
- Strategies for enhancing signal quality and accuracy
- Future directions enabled by multisensory interfaces

6 Practical Work

We will have exercises in jupyter notebooks/google colab, walk-throughs of implementing multimodal AI systems, and video + demos on latest research in multimodal AI and sensing.

7 Instructor Backgrounds

The 3 instructors provide complementary expertise in AI, HCI, sensing, and hardware.

Paul Liang is an Assistant Professor at the MIT Media Lab and MIT EECS. He is the founding director of the Multisensory Intelligence research group, which studies the foundations of multi-sensory artificial intelligence to create human-AI symbiosis across scales and sensory mediums, enhancing productivity, creativity, and wellbeing. He is a recipient of the Siebel Scholars Award, Waibel Presidential Fellowship, Facebook PhD Fellowship, Center for ML and Health Fellowship, Rising Stars in Data Science, and 3 best paper awards. Outside of research, he received the Alan J. Perlis Graduate Student Teaching Award for developing new courses and tutorials on multimodal machine learning.

Karan Ahuja is the Lisa Wissner-Slivka and Benjamin Slivka Assistant Professor in Computer Science at Northwestern University where he directs the Sensing, Perception, Interactive Computing and Experiences (SPICE) Lab. His research group creates cutting-edge computing technologies that sense, track, and understand humans to augment their interactions and assist them in daily life. He received his Ph.D. in Human-Computer Interaction from Carnegie Mellon University in 2023 and B.Tech. in Computer Science in 2017. To date, Karan has published over 30 papers, with 7 winning awards, at top human-computer interaction venues (ACM UIST, CHI, UbiComp among others). Karan is also a recipient of the Forbes 30 under 30, ACM SIGCHI Outstanding Dissertation Award, and Siebel Fellowship.

Yiyue Luo is an Assistant Professor of Electrical and Computer Engineering at the University of Washington, directing the Wearable Intelligence Lab. She received her PhD and Master degrees in EECS from MIT. Her research lies at the intersection of digital fabrication, human-computer/robot interaction, and applied AI. Her research on integrated intelligent textiles has been published in interdisciplinary journals, e.g., *Nature Electronics* and *Nature Communications*, top human-computer interactions, robotics and learning venues, e.g., CHI, UIST, CVPR, ICRA. She has been awarded fellowship from MathWorks, Google, and Accenture, and was listed as Forbes 30 under 30 North American 2024.

8 Resources

Paul Liang co-created and taught the annual Multimodal Machine Learning course at CMU (course 11-877 and 11-777), he is now creating and starting to teach the annual Multimodal AI course at MIT. For recent iterations of the course and tutorials at international conferences, the materials and videos are publicly available at:

- <https://cmu-multicomp-lab.github.io/adv-mmml-course/spring2023/>
- <https://cmu-multicomp-lab.github.io/mmml-course/fall2022/>

- ICML 2023: 300 attendees. 4-hour tutorial, slides and videos available at: <https://cmu-multicomp-lab.github.io/mmml-tutorial/icml2023/>.
- CVPR 2022: 300 attendees. 4-hour tutorial, slides and videos available at: <https://cmu-multicomp-lab.github.io/mmml-tutorial/cvpr2022/>.
- NAACL 2022: 200 attendees. 4-hour tutorial, slides and videos available at: <https://cmu-multicomp-lab.github.io/mmml-tutorial/naacl2022/>.

Since they were presented at ICML/CVPR/NAACL, they are geared toward the core AI and ML communities, with a greater focus on models and applications. Our proposed CHI 2025 course will shorten the foundations of multimodal AI into the first 45 minutes, and add 2 significant HCI components: first on human sensing for 45 minutes and the second on hardware for multimodal AI for 45 minutes.

Karan Ahuja has taught the Machine Learning and Sensing course at CMU (17-728) and is currently teaching the Advanced Topics in Ubiquitous Computing course at Northwestern University, link here: <https://www.mccormick.northwestern.edu/computer-science/academics/courses/descriptions/396-496-21.html>

Yiyue Luo has hosted workshops on intelligent wearable technologies at ICRA 2023, ACM UIST 2024, and UbiComp 2024. She is teaching the Research Seminar on Wearable Intelligence at the University of Washington, link here: <https://advisingblog.ece.uw.edu/2024/06/04/ece-ee-539c-research-seminar-on-wearable-intelligence-aut-24/>

Our course is also related to the following previous tutorials at HCI conferences:

- Solving the Activity Recognition Problem: Self-supervised and multi-modal recognition of activities from wearable sensors (UbiComp'23): <https://sites.google.com/view/soar-tutorial-ubicomp2023/home>
- Deep Learning for Human Activity Recognition (UbiComp'21): <https://www.ubicomp.org/ubicomp2021/program/virtual-tutorials/>

Our course differs from these because we cover start-of-the-art AI principles instead of focusing only on applications of AI in human sensing.

9 Accessibility

We will ensure accessibility of our materials to audiences with diverse needs by including captioning, providing accessible PDFs that are compatible with screen readers, and offering alternative formats for visual or auditory content upon request. We will require only basic AV equipment suitable for lectures from the organization committee.

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